

Notes: Paravisini (JF, 2008)
Local Bank Financial Constraints and Firm Access to External Finance

Contents

1	Big picture	2
2	Incremental contribution of the paper	2
3	Data and measurement	3
3.1	The MYPES program	3
3.2	Bank-level financial statements	4
3.3	Loan-level credit registry data	4
3.4	Expected external financing instrument	4
3.5	Outcome and quality measures	5
4	Models and empirical specifications	5
4.1	Bank lending sensitivity to external finance	5
4.2	Program allocation and expected financing	6
4.3	First-stage regression	6
4.4	Lagged lending response	6
4.5	Underinvestment versus free cash flows	7
4.6	Borrower-level debt response	7
5	Identification and empirical design	8
5.1	What the design can identify	8
5.2	Relevance of the instrument	8
5.3	Exclusion restriction	8
5.4	Endogenous bank self-selection	8
5.5	Targeting and loan relabeling	9
5.6	Underinvestment interpretation	9
5.7	Limitations of the design	9
6	Tables and figures	10
6.1	Figure 1 and Figure 2: Loan relabeling and program waves	10

6.2	Table I and Table II: Bank and loan summary statistics	11
6.3	Figure 3 and Table III: Default timing and first stage	12
6.4	Table IV and Table V: Lending response to external financing	13
6.5	Table VI and Table VII: Profitability and borrower access to finance	14
6.6	Table VIII and Table IX: Testing identifying assumptions	15
7	Short summary	16
8	Conclusion	17
8.1	Core conclusion	17
8.2	Economic intuition	17
8.3	Incremental contribution revisited	17
8.4	Implications for banks, borrowers, and policy	17
8.5	Limitations	17
8.6	Takeaway	18

1 Big picture

- **Question.** Do financing frictions at local banks affect the supply of credit to firms? More specifically, when a constrained bank receives additional external financing, does it expand lending, and does this expansion reflect profitable underinvestment rather than wasteful free-cash-flow lending?
- **Setting.** The paper studies Argentina's banking system around the MYPES government on-lending program, which allocated external funds to banks in waves between 1993 and 1999 using a formula based on bank characteristics.
- **Core identification idea.** The paper exploits the exogenous component of the MYPES allocation formula. Wave size, interacted with predetermined bank scores based on average loan size and lending in poor provinces, predicts program financing but is argued to be unrelated to bank-specific credit demand or marginal investment opportunities.
- **Main bank-level result.** Banks expand lending strongly when external financing rises. The preferred 2SLS estimate implies that loans increase by about \$0.66 for each additional dollar of external financing immediately, and the cumulative expansion averages more than \$1.2 per dollar over the following two years.
- **Profitability result.** Loans made during the externally financed lending expansions do not have higher default rates, and interest revenue per dollar of loans does not fall. This supports an underinvestment interpretation rather than a free-cash-flow interpretation.
- **Borrower-level result.** Borrowers' total bank debt increases when their lenders receive financing shocks, even after borrower fixed effects and within-firm comparisons. This implies that other banks do not fully arbitrage the constrained bank's forgone lending opportunities.
- **Mechanism.** Local banks appear financially constrained. Relaxing the constraint allows them to fund profitable loans that were previously not supplied.
- **Aggregate implication.** Financial constraints at local intermediaries can reduce aggregate credit availability. This matters even in a banking system with substantial foreign-bank presence.
- **Economic takeaway.** The paper provides direct evidence that intermediary financial constraints can create real credit rationing for firms. Bank balance-sheet shocks are not neutral: they move lending quickly, persistently, and with amplification.

2 Incremental contribution of the paper

- **From bank lending correlations to bank-credit supply shocks.** Earlier lending-channel papers often use deposits, monetary policy, or bank liquidity as proxies for external finance. This paper uses a formula-based government financing allocation to obtain plausibly exogenous variation in bank funding.
- **A direct test of constrained intermediaries.** The paper does not only ask whether bank lending is correlated with bank liquidity. It estimates the causal sensitivity of lending to an external-finance shock, holding macro conditions and time-invariant bank heterogeneity fixed.

- **A three-step underinvestment test.** The paper explicitly tests three hypotheses: banks are financially constrained; financing constraints prevent profitable lending; and constrained-bank borrowers cannot fully replace lost credit from other lenders.
- **A marginal-loan quality contribution.** Using loan-level credit registry data, the paper measures ex post default performance of loans issued during lending expansions. This is a direct way to distinguish underinvestment from value-destroying risk taking.
- **A borrower-level substitution contribution.** The paper shows that total borrower debt rises when the borrower bank receives external financing. This goes beyond a bank-level lending result and speaks to whether credit supply is reallocated by competing banks.
- **A formula-instrument contribution.** The empirical strategy turns a bureaucratic allocation rule into an instrument. This makes the paper methodologically similar to natural-experiment work in corporate finance, but applied to local bank balance sheets.
- **A policy contribution.** The results imply that improving local banks' access to capital markets can have first-order effects on credit allocation, especially where relationship lending and information frictions make local-bank credit hard to substitute.
- **A financial-intermediation contribution.** The paper provides evidence that agency and information frictions between capital providers and intermediaries can generate underinvestment, not just distortions inside nonfinancial firms.

3 Data and measurement

3.1 The MYPES program

- The Credit Program to Small and Medium Sized Firms, or MYPES, was implemented in Argentina between 1993 and 1999.
- The program provided banks with three-year financing at the average deposit rate in the banking system.
- MYPES was funded by the Inter-American Development Bank and was intended to encourage financial intermediaries to lend to small businesses.
- Eligible firms had fewer than 20 workers and less than \$200,000 in annual sales.
- Banks were supposed to lend \$1 to eligible borrowers for each \$0.75 of program financing received, but the paper shows that the targeting rule was not binding because banks could relabel existing eligible loans as program loans.
- Program financing was allocated in 12 waves. The paper's preferred identification comes from the final four waves, which were compressed into 1998-1999 because of an administrative deadline rather than banking-sector liquidity needs.

Interpretation: MYPES creates a financing shock to banks. The key empirical advantage is that the amount of financing is partly driven by an administrative formula and wave timing, not by each bank's own contemporaneous investment opportunities.

3.2 Bank-level financial statements

- The bank data contain monthly balance sheets and earnings reports for nongovernment banks in Argentina between January 1995 and December 2001.
- The main bank-level outcome is lending growth, measured using changes in log loans.
- The key financing variable is sources of capital: equity, deposits, and other liabilities minus regulatory reserve requirements.
- The preferred sample period is January 1998 through December 2000, because it covers the final four MYPES waves and the post-shock lending response window.
- The paper controls for bank fixed effects and month fixed effects, and in some specifications uses bank-level controls such as assets, deposits, cash flows, and net income.

Interpretation: The bank panel allows the paper to isolate within-bank changes in lending around externally induced funding changes while absorbing common macroeconomic shocks.

3.3 Loan-level credit registry data

- The public credit registry, CDSF, records monthly loan-level information for firms and individuals with more than \$50 of debt with a financial institution in Argentina.
- Each loan observation contains debtor ID, bank ID, principal outstanding, collateral, and repayment-status codes.
- The repayment-status codes allow the paper to measure whether loans issued during financing expansions later become nonperforming.
- The preferred loan sample covers loans issued between January 1998 and December 2000.
- The data allow the paper to distinguish existing borrowers from new borrowers and to aggregate borrowers' total bank debt across all lenders.

Interpretation: The credit-registry data make the paper much stronger than a standard bank balance-sheet study. They allow a test of marginal loan quality and a borrower-level test of credit substitution.

3.4 Expected external financing instrument

- Actual program financing may be endogenous because banks choose whether to participate.
- The paper therefore constructs expected program financing, $\hat{C}_{it}$, using the predicted probability of participation and the administrative allocation formula.
- The allocation formula gives greater weight to banks with smaller average loans and more lending in poor provinces.
- The instrument varies over time through wave size and across banks through predetermined formula scores.

- The final four waves are central because their timing was caused by an administrative rush to spend the remaining MYPES budget.

Interpretation: The instrument is not just program participation. It is the part of program funding predicted by mechanical formula exposure and wave-level resources.

3.5 Outcome and quality measures

- Bank lending is measured as growth in bank loans.
- Bank financing is measured as growth in sources of capital.
- Loan risk is measured by whether a newly issued loan becomes nonperforming within 12 or 24 months.
- Loan return is proxied by interest revenue per dollar of loans and net interest revenue per dollar of loans.
- Borrower access to finance is measured as growth in the borrower's total bank debt across all banks.

Interpretation: The paper is careful to distinguish quantity from quality. A lending expansion alone would not prove underinvestment; the risk and return evidence is needed to rule out wasteful lending.

4 Models and empirical specifications

4.1 Bank lending sensitivity to external finance

The baseline bank-level relationship is

$$\ln L_{it} - \ln L_{it-1} = \alpha_i + \alpha_t + \beta_0(\ln F_{it} - \ln F_{it-1}) + x_{it}\gamma_1 + \varepsilon_{it},$$

where L_{it} is loans of bank i at month t and F_{it} is sources of capital.

- β_0 is the key lending-sensitivity parameter.
- If banks are unconstrained and program financing is inframarginal, extra financing should be used to reduce other liabilities or pay investors, not to expand lending.
- A positive β_0 therefore rejects the null of no financing frictions.
- The paper estimates β_0 by 2SLS, using changes in expected external financing as the instrument for changes in sources of capital.

Interpretation: The empirical test is analogous to an investment-cash-flow sensitivity test, but the shock is to bank external finance and the outcome is credit supply.

4.2 Program allocation and expected financing

Actual program financing in wave w is allocated by formula:

$$C_{iw} = A_w \left[\frac{Zregion_i}{2 \sum_j^{n_w} Zregion_j} + \frac{Zsize_i}{2 \sum_j^{n_w} Zsize_j} \right],$$

where A_w is the wave size, $Zregion_i$ is the score based on lending in poor regions, and $Zsize_i$ is the score based on average loan size.

Because participation is endogenous, the instrument uses predicted participation probabilities:

$$\hat{C}_{iw} = A_w \left[\frac{\hat{p}_{iw} Zregion_i}{2 \sum_j^N \hat{p}_{jw} Zregion_j} + \frac{\hat{p}_{iw} Zsize_i}{2 \sum_j^N \hat{p}_{jw} Zsize_j} \right].$$

- The predicted probability $\hat{p}_{iw}$ comes from a participation model using wave size and predetermined bank formula scores.
- The expected allocation uses all potential participants, not only the self-selected set of actual participants.
- This is designed to purge the endogenous component of bank participation.

Interpretation: The key source of identifying variation is that a large wave mechanically matters more for banks with higher formula exposure.

4.3 First-stage regression

The first-stage regression is

$$\ln F_{it} - \ln F_{it-1} = \alpha_i + \alpha_t + \varphi(\ln \hat{C}_{it} - \ln \hat{C}_{it-1}) + x_{it}\gamma_2 + \eta_{it}.$$

- The first stage tests whether expected program financing actually raises bank sources of capital.
- The coefficient φ is positive and significant.
- The preferred estimate implies that bank sources of capital rise roughly one-for-one with program financing.

Interpretation: The first stage validates that the formula-based instrument is relevant and that program funding did not simply replace other sources of finance.

4.4 Lagged lending response

To measure persistence and amplification, the paper estimates

$$\begin{aligned} \ln L_{it} - \ln L_{it-1} = & \alpha_i + \alpha_t + \beta_0(\ln F_{it} - \ln F_{it-1}) \\ & + \beta_1(\ln F_{it-1} - \ln F_{it-2}) + \dots + \beta_{24}(\ln F_{it-24} - \ln F_{it-25}) + \varepsilon_{it}. \end{aligned}$$

- Each lagged financing change is instrumented with the corresponding lag of expected external financing.

- The cumulative effect sums the marginal coefficients over time.
- The lending response is quick, persistent, and larger than the initial dollar shock.

Interpretation: The lag structure is important because constrained banks may take time to identify borrowers, process loans, and generate additional deposit relationships.

4.5 Underinvestment versus free cash flows

The loan-level default test is

$$Default_{ijt}^T = \alpha_i + \alpha_t + \alpha_s + \psi_T(\ln F_{it} - \ln F_{it-1}) + \omega_{ijt},$$

where $Default_{ijt}^T$ equals one if loan j issued by bank i becomes nonperforming within T months.

The bank-level return test is

$$Interest\ Revenue_{it}/Loans_{it} = \alpha_i + \alpha_t + \gamma(\ln F_{it} - \ln F_{it-1}) + \omega_{it}.$$

- Under the free-cash-flow story, extra financing should push banks into lower-quality or lower-return loans.
- Under the underinvestment story, the marginal loans should still be profitable.
- The paper finds no increase in default rates and no decline in interest revenue per dollar of loans.

Interpretation: This is the paper's main test that bank financial constraints are inefficient rather than simply disciplining managers.

4.6 Borrower-level debt response

The firm-level debt regression is

$$\ln Debt_{kt} - \ln Debt_{kt-1} = \alpha_k + \alpha_t + \Delta_F(\ln F_{kt} - \ln F_{kt-1}) + \nu_{kt},$$

where F_{kt} averages sources of capital across firm k 's lenders.

The within-firm bank-borrower specification is

$$\begin{aligned} \ln Debt_{ikt} - \ln Debt_{ikt-1} = & \alpha_k + \alpha_i + \alpha_t + \Delta_L(\ln F_{ikt} - \ln F_{ikt-1}) \\ & + \chi(\ln Debt_{-ikt} - \ln Debt_{-ikt-1}) + v_{ikt}. \end{aligned}$$

- The firm-level specification asks whether total borrower debt rises when its lenders receive funding shocks.
- The within-firm specification compares borrowing from the shocked bank with borrowing from the same firm's other banks.
- The positive estimates imply that borrowers could not fully replace constrained-bank credit elsewhere.

Interpretation: This is the bridge from bank constraints to firm access to external finance.

5 Identification and empirical design

5.1 What the design can identify

- The paper identifies a local causal effect of externally induced bank financing on lending for banks exposed to the MYPES allocation formula.
- It is strongest for the final four program waves, whose timing reflects an administrative deadline rather than banking-sector liquidity needs.
- The estimates should be interpreted as applying to local banks in an emerging-market banking system where relationship lending and informational frictions are important.

Interpretation: The design is not a randomized experiment, but the formula and administrative timing create quasi-experimental variation in bank funding.

5.2 Relevance of the instrument

- Table III shows that expected external financing predicts changes in bank sources of capital.
- The relationship is strong for banks with a high predicted probability of participation.
- It is weak or absent for low-probability banks, which supports the view that the instrument operates through program exposure rather than a general macro factor.

Interpretation: The first stage is economically meaningful: program money increases available bank capital instead of crowding out other financing one-for-one.

5.3 Exclusion restriction

- The identifying assumption is that expected external financing is unrelated to unobserved changes in bank investment opportunities or marginal cost of capital.
- Bank fixed effects absorb time-invariant bank characteristics.
- Month fixed effects absorb common macroeconomic and banking-sector shocks.
- The final-four-wave analysis helps avoid the concern that program waves were timed to coincide with banking-sector liquidity stress.

Interpretation: The paper's exclusion restriction is plausible because the identifying variation comes from formula exposure interacted with administrative wave size, not from banks choosing when they need funds.

5.4 Endogenous bank self-selection

- Banks with liquidity needs or good lending opportunities may self-select into the program.
- The paper addresses this by predicting participation using wave size and predetermined formula scores.

- Expected allocation is then calculated using predicted participation probabilities rather than actual self-selected participation.

Interpretation: The instrument attempts to keep the mechanical part of program allocation and remove the discretionary participation part.

5.5 Targeting and loan relabeling

- MYPES formally targeted small firms, but the monitoring system allowed banks to relabel existing loans to eligible borrowers as program loans.
- Figure 1 shows that total debt from the program bank does not increase for the firms receiving program loans; instead, program debt replaces other debt from the same bank.
- Later results show that the lending expansion is mainly allocated to noneligible firms.

Interpretation: These facts weaken the concern that the estimated lending response is mechanically caused by a binding program rule requiring new small-firm lending.

5.6 Underinvestment interpretation

- A positive lending response alone could reflect either underinvestment or managerial free cash flow.
- The paper rules against free cash flow by showing that marginal loans made during funding expansions are not riskier and do not generate lower interest revenue.
- The borrower-debt results show that constrained-bank borrowers were credit rationed, not merely switching lenders.

Interpretation: The empirical design is layered: first show bank financing constraints, then show marginal loans are not worse, and finally show borrowers' total debt increases.

5.7 Limitations of the design

- The setting is Argentina in the 1990s, so external validity to developed-market banking systems is not automatic.
- The preferred sample immediately precedes the 2001 Argentine crisis, although the paper argues the banking sector remained relatively healthy before the crisis.
- Default rates are measured during normal times and cannot fully reveal marginal-loan performance under severe aggregate stress.
- The instrument identifies effects for banks affected by MYPES, not necessarily all banks.

Interpretation: The paper is compelling for the local setting and for the mechanism it studies, but its estimates should be generalized with care.

6 Tables and figures

6.1 Figure 1 and Figure 2: Loan relabeling and program waves

Read:

- Figure 1 plots monthly debt around the date firms receive MYPES program loans.
- Program debt rises at month 0, but other debt from the same program bank falls almost one-for-one.
- This pattern indicates loan relabeling: banks can satisfy program paperwork without necessarily extending new credit to the formally targeted firms.
- Figure 2, Panel A, shows annual program resources and participating banks. Program financing peaks during the mid-1990s and again in 1999.
- Figure 2, Panel B, shows the 12 program waves. Waves 9 through 12 occur between December 1998 and November 1999.
- The final four waves are important because they reflect an administrative rush to spend remaining funds.

Economic message: Figure 1 supports the claim that program targeting was not binding, while Figure 2 motivates the timing-based identification strategy. The paper uses the final administrative waves to obtain cleaner variation in bank external finance.

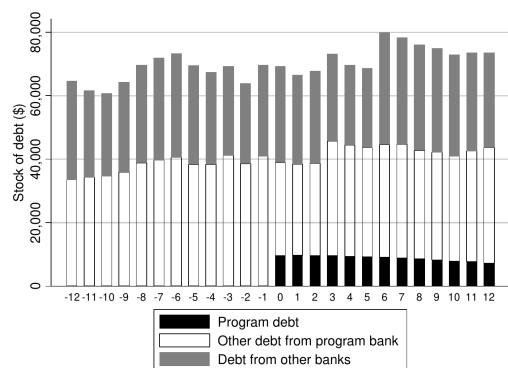


Figure 1. Evidence on loan re-labeling: monthly debt evolution of the firms that receive program loans, by source. Monthly evolution of total bank debt of a sample of firms that receive MYPES loans, classified by source. Calculations are based on a sample of 2,596 firms that receive program loans after January 1996. The horizontal axis measures time in months relative to the date each firm received the MYPES loan.

a Figure 1: Evidence on loan relabeling

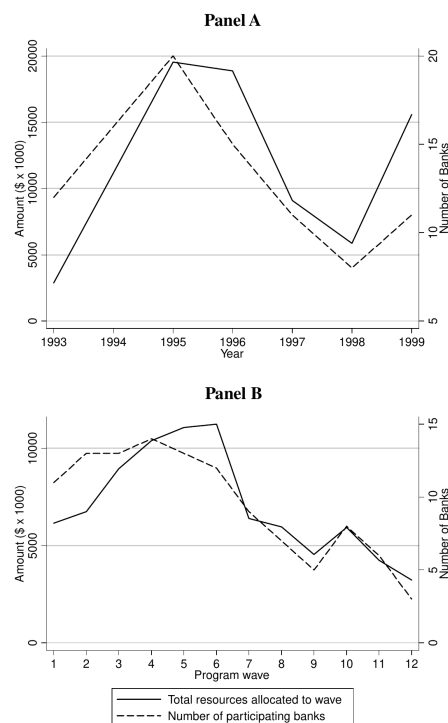


Figure 2. Panel A: Flow of program financing and number of participating banks year. The flow of financing during a year is the sum of the amount of resources allocated to all waves that began during that year. The number of banks counts a bank once if it participates in two waves during the same year. Panel B: Flow of program financing and number of participating banks, by wave. Waves 9 through 12 occurred between December 1998 and November 1999.

b Figure 2: Program financing and participating banks

6.2 Table I and Table II: Bank and loan summary statistics

Read:

- Table I describes nongovernment banks by ownership and program participation.
- Program banks are smaller than the average bank but still economically meaningful, with average assets of about \$544 million and loans of about \$284 million.
- Expected external financing represents about 7.6% of loans for program banks.
- Table II describes the loan-level sample. The preferred sample contains 750,526 loans, commitments, or credit-limit increases.
- Existing borrowers receive most loans, but new borrowers have higher default rates: 19.1% default after 12 months and 22.8% after 24 months.
- The average loan amount is \$16,691 and collateral covers about 12.3% of loan principal.

Economic message: These tables establish the empirical scale. MYPES financing is small relative to the whole banking system but material for participating banks, and the loan-level data are rich enough to study marginal loan risk.

Table I
Bank Descriptive Statistics (Nongovernment Banks), by Ownership and Program Participation (Thousands of \$)

Means and standard deviations (in brackets) are reported. The statistics are calculated for a sample of 111 banks (83 locally owned, 26 of which received external financing from the program at some time between 1993 and 1999), first averaging each variable across all monthly observations between January 1998 and December 2000 (the preferred sample period) and then across banks. The reported standard errors are calculated from the cross-sectional variation, calculated after averaging each variable across monthly observations between January 1998 and December 2000. Sources of capital: the sum of equity, deposits, and other liabilities minus the reserve requirements for each type of liability (for example, 20% for checking accounts, 5% for 90-day deposits, 0% for 1-year deposits, and so on). Expected External Financing is the amount of financing a bank expects to receive from the MYPES program, and is calculated using the estimation of a participation model and the cross-sectional formula. Program banks hold on average 10.2%, and local banks 51% of the total assets of the banking system.

	All Banks (1)	Locally Owned Banks (2)	Program Banks (3)
Assets	1,095,287 [2,396,431]	693,538 [1,880,620]	543,985 [599,719]
Loans	536,344 [1,241,008]	362,000 [1,043,741]	283,790 [332,044]
Loans and commitments	636,382 [1,541,574]	584,056 [1,751,396]	345,905 [394,131]
Liabilities	979,350 [2,168,293]	605,314 [1,674,728]	488,200 [539,852]
Deposits	569,590 [1,331,549]	384,378 [1,122,679]	361,719 [407,961]
Sources of capital	616,099 [1,348,276]	399,500 [1,151,055]	382,853 [418,375]
Loans/assets	0.500 [0.146]	0.517 [0.140]	0.500 [0.109]
Deposits/assets	0.515 [0.194]	0.538 [0.189]	0.626 [0.124]
Equity/assets	0.133 [0.135]	0.144 [0.141]	0.133 [0.130]
ROA	0.31% [1.22]	0.26% [1.23]	0.14% [1.12]
Financial rev/loans (%)	13.6% [7.2]	14% [7.6]	12.8% [2.4]
Exp. external financing	1,547.5 [494.9]	1,630.4 [482.4]	1,598.7 [429.2]
Exp. ext. financing/loans	0.068 [0.166]	0.068 [0.154]	0.076 [0.196]

a Table I: Bank descriptive statistics

Table II
Loan and Loan Recipient Summary Statistics, by Borrower Type

Means and standard deviations (in brackets) are reported. Statistics are estimated using the January 1998 to December 2000 preferred subsample. Each observation corresponds to a new loan, loan commitment, or increase in credit limit extended during the sample period. Loan amount represents the amount of the loan or credit availability expansion. Default after 12 (24) months is a dummy equal to one if the loan is nonperforming 12 (24) months after the loan is issued. Total bank debt is the sum of the bank debt held by the loan recipient across all financial institutions. Past nonperforming loan is a dummy equal to one if the loan recipient has any non-performing debt at the time of receiving the new loan. A loan recipient is classified as *new* if it has no previous credit with the issuing bank, and *existing* otherwise.

	All (1)	Existing Borrowers (2)	New Borrowers (3)
Number of Loans	750,526	620,325	130,201
1. Loan characteristics			
Loan amount (\$)	16,691 [226,660]	17,722 [218,790]	11,776 [260,858]
Collateral/loan	0.123 [0.301]	0.124 [0.299]	0.118 [0.312]
2. Loan performance			
Default after 12 months (yes = 1)	0.122 [0.328]	0.104 [0.306]	0.191 [0.393]
Default after 24 months (yes = 1)	0.168 [0.374]	0.153 [0.359]	0.228 [0.420]
3. Borrower history			
Total bank debt		58,551 [601,468]	
Past non-performing loan (yes = 1)		0.141 [0.348]	

b Table II: Loan and borrower summary statistics

6.3 Figure 3 and Table III: Default timing and first stage

Read:

- Figure 3 plots the monthly hazard rate of default for loans with at least three years of repayment history.
- The default hazard peaks before 12 months and declines afterward.
- This supports using 12- and 24-month default windows to measure loan performance.
- Table III is the first stage. Expected external financing predicts growth in bank sources of capital.
- The coefficient is 0.068 without controls and 0.076 with additional controls, both statistically significant.
- The first stage is present for banks with high predicted participation probability and absent for low-probability banks.

Economic message: Figure 3 validates the default-window measurement, while Table III validates the instrument. The expected financing variable shifts bank balance-sheet capacity in the intended direction.

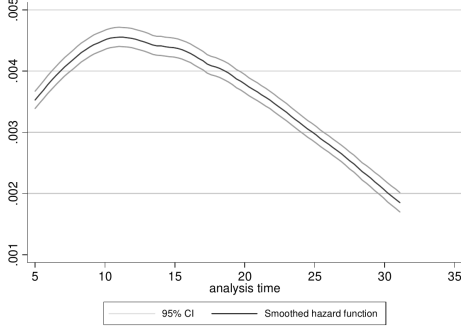


Figure 3. Kernel estimation of the monthly hazard rate of default. The vertical axis is the weighted kernel density estimate (smoothed) of the monthly hazard function of defaults—probability that a loan issued at month t defaults at month $t + T$, conditional on not having defaulted at $t + T - 1$. The horizontal axis measures months elapsed after the loan was issued. The kernel is estimated over the subsample of loans issued between January and August 1998, with at least 12 years of observable repayment history.

a Figure 3: Monthly hazard rate of default

Table III
First Stage: Regression of Sources of Capital on Expected External Financing and Bank/Month Fixed Effects

The table presents the estimated coefficients of the following specification:

$$\ln F_{it} - \ln F_{it-1} = \alpha_i + \alpha_t + \varphi(\ln \hat{C}_{it} - \ln \hat{C}_{it-1}) + \mathbf{x}_{it}'\gamma_2 + \eta_{it},$$

where F_{it} represents sources of capital (sum of equity, deposits, and other liabilities minus the reserve requirements for each type of liability) of bank i at time t , $\hat{C}_{it}$ is the expected external financing, α_i and α_t are bank and month fixed effects, and $\mathbf{x}_{it}$ is a set of bank-level controls (lagged log of assets, deposits, cash flows, and net income). All estimates are based on the sample months between January 1998 and December 2000, which includes the final four waves of the program. The subsample partition in columns 2 and 3 is based on the predicted probability of participation in MYPES. Banks are defined as having a high average probability of participation if they are in the top quartile of the distribution estimated participation probabilities. Heteroskedasticity-robust standard errors are in brackets, clustered at the bank level. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

Bank Subsample	All		High Probability of Participation	Low Probability of Participation
	(1)	(2)	(3)	(4)
Dependent Variable: Bank Sources of Capital (logs, first differences)				
Expected external financing (logs, first differences)	0.068*** [0.021]	0.076*** [0.025]	0.067** [0.030]	0.016 [0.024]
Additional controls	No	Yes	No	No
Bank/month FE	Yes	Yes	Yes	Yes
Observations	4,654	4,654	1,405	3,493
R-squared	0.07	0.08	0.06	0.04

b Table III: First-stage regression

6.4 Table IV and Table V: Lending response to external financing

Read:

- Table IV estimates the instantaneous effect of sources of capital on bank loans.
- The preferred final-four-wave 2SLS estimate is 0.745, larger than the OLS estimate of 0.314.
- The coefficient implies roughly \$0.66 more lending for each additional dollar of external finance at sample averages.
- Using all waves produces a smaller 2SLS estimate, consistent with early waves being more confounded by banking-sector liquidity stress.
- Table V adds lagged effects for up to 24 months.
- The cumulative effect is positive and economically large; the average cumulative elasticity is around 1.3, implying more than \$1.2 of lending per dollar of financing.

Economic message: The core lending result is quick, persistent, and amplified. A local bank's external-finance shock has a large effect on its credit supply, inconsistent with frictionless Modigliani-Miller behavior for banks.

Table IV

OLS/2SLS Estimates of the Instantaneous Effect of an Expansion in Bank External Financing on Loans, by Bank Size and Capitalization

Each column presents estimated coefficients from a specification of the form:

$$\ln L_{it} - \ln L_{it-1} = \alpha_i + \alpha_t + \beta_0(\ln F_{it} - \ln F_{it-1}) + \beta_0^c(\ln F_{it} - \ln F_{it-1}) \text{Constrained Bank}_i + \varepsilon_{it},$$

where L_{it} are loans of bank i at month t , F_{it} are sources of capital (total capital minus regulatory reserve requirements), *ConstrainedBank_i* is a dummy equal to one if bank i is likely to be financially constrained according to an ex ante observable characteristic, and α_i and α_t are bank and month fixed effects. As constrained banks indicators, specification (3) uses *Small_{it}*, a dummy equal to one if the bank is in the lowest quartile of the asset distribution, and specification (4) uses *LowCap_{it}*, a dummy equal to one if the bank is in the lowest quartile of the equity to assets ratio distribution. Both variables are predetermined, calculated at the time the banks first appear in the sample. Of the program banks in the 1998–2000 sample, 40.7% are classified as small and 25.9% as low capitalized. Of the nonprogram banks, 51.0% are classified as small and 14.9% as low capitalized. The 2SLS estimates are obtained using the change in expected external financing as an instrument for the change in bank sources of capital, and expected external financing interacted with *Small_{it}* and *LowCap_{it}* as an instrument for the interaction term between sources of capital and the dummy variables. The first stage for all estimates is shown in Table III. The reduced sample of specifications 3 through 6 covers years 1998 and 1999, which include the last four waves of the program. Heteroskedasticity-robust standard errors are in brackets, clustered at the bank level. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

Estimation Method	All Waves		Final Four Waves			
	OLS (1)	2SLS (2)	OLS (3)	2SLS (4)	2SLS (5)	2SLS (6)
Dependent Variable: Loans (logs, first differences)						
Sources of capital (logs, first differences)	0.309*** [0.074]	0.474* [0.260]	0.314*** [0.090]	0.745*** [0.139]	0.618*** [0.171]	0.708*** [0.174]
Sources of capital × Small				0.065 [0.211]		
Sources of capital × LowCap					0.105 [0.204]	
Bank/month FE	Yes	Yes	Yes	Yes	Yes	Yes
# Banks	119	119	113	113	113	113
Observations	6,330	6,330	4,654	4,654	4,654	4,654
R-squared	0.14		0.17			

a Table IV: Instantaneous lending response

Table V

2SLS Estimates of the Lagged Effects of an Expansion in Bank External Financing on Loans

The marginal lagged effects are the 2SLS estimation results of the coefficients of the following specification:

$$\ln L_{it} - \ln L_{it-1} = \alpha_i + \alpha_t + \beta_0(\ln F_{it} - \ln F_{it-1}) + \beta_1(\ln F_{it-1} - \ln F_{it-2}) + \beta_2(\ln F_{it-2} - \ln F_{it-3}) + \dots + \beta_T(\ln F_{it-24} - \ln F_{it-25}) + \varepsilon_{it},$$

where L_{it} are loans of bank i at month t , F_{it} are sources of capital (total capital minus regulatory reserve requirements), and α_i and α_t are bank and month fixed effects. The specification is estimated including all lags, but only the coefficients for every quarter are reported for conciseness. The 2SLS estimates are obtained using the change in expected external financing and its lags as an instrument for the change in bank sources of capital. The cumulative lagged effects up to time s are the sum of the marginal coefficients β_0 through β_s . The estimations are obtained from the reduced sample that includes the last four waves of the program. Heteroskedasticity-robust standard errors, clustered at the bank level, are reported. Both specifications include a full set of bank and month dummies. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

Lags in Months	Marginal Lagged Effect		Cumulative Lagged Effect	
	Point Estimate (1)	SE (2)	Point Estimate (3)	SE (4)
0	0.887*	0.476	0.887*	0.476
3	0.475	0.309	1.303*	0.690
6	−0.270	0.306	0.747	0.634
9	−0.265	0.470	1.434*	0.845
12	0.178	0.282	0.935	0.610
15	0.226	0.489	0.954*	0.557
18	0.319	0.509	1.996*	1.156
21	0.198	0.515	2.376*	1.429
24	−0.200	0.263	1.300	1.865

b Table V: Lagged lending response

6.5 Table VI and Table VII: Profitability and borrower access to finance

Read:

- Table VI asks whether the new lending is worse lending.
- Loans issued during external-finance expansions are not more likely to become nonperforming after 12 or 24 months.
- Interest revenue over loans does not decline, and net interest revenue over loans increases.
- These results are inconsistent with a free-cash-flow story in which managers use extra funds to make unprofitable loans.
- Table VII asks whether borrowers' total bank debt increases when their lenders receive financing shocks.
- The positive borrower-debt coefficients imply that constrained-bank borrowers expand total borrowing rather than simply replacing one lender with another.
- The effect is driven by noneligible firms, which further supports the claim that the formal MYPES targeting rule is not driving the lending expansion.

Economic message: These two tables are the heart of the underinvestment interpretation. The marginal loans do not look lower quality, and borrowers appear credit constrained, so local bank financing frictions reduce real access to external finance.

Table VI
Constraints or Free Cash Flows? 2SLS Estimates of the Effect of an Expansion in Bank External Financing on Loan Default Rates and Interest Revenues

Columns 1 and 2 present the estimated coefficients from a specification of the form:

$$DefaultT_{ijt} = \alpha_i + \alpha_t + \alpha_s + \psi_T(\ln F_{it} - \ln F_{it-1}) + \omega_{ijt},$$

where each observation corresponds to a loan j issued by bank i at month t . $DefaultT_{ijt}$ is a dummy equal to one if loan j is nonperforming (at least 6 months late, defaulted, or the loan recipient has filed for bankruptcy) at some time between time t and time $t + T$, where T can be 12 or 24 months (specifications (1) and (2), respectively). F_{it} are sources of capital (total capital minus regulatory reserve requirements). Finally, α_i , α_t , and α_s are bank, month, and loan recipient sector (three-digit industry sector) fixed effects.

Columns 3 and 4 present the estimated coefficients from a bank-level specification of the form:

$$Interest\ Revenue_{it}/Loans_{it} = \alpha_i + \alpha_t + \gamma(\ln F_{it} - \ln F_{it-1}) + \omega_{it}.$$

The 2SLS estimates are obtained using the change in expected external financing as an instrument for the change in bank sources of capital. Heteroskedasticity-robust standard errors are in brackets, clustered at the bank level. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

	Loan Non-Performing After		Interest Revenue/Loans	Net Interest Revenue/Loans
	12 Months (1)	24 Months (2)		
1. Sample: all loans				
Sources of capital	0.0007	-0.0005	0.004	0.033*
(logs, first differences)	[0.0015]	[0.0011]	[0.032]	[0.018]
Bank/month fixed effects	Yes	Yes	Yes	Yes
Industry Fixed effects	Yes	Yes	No	No
Observations	750,563	750,563	4,654	4,654
2. Sample: existing borrowers				
Sources of capital	-0.0001	-0.0013		
(logs, first differences)	[0.0012]	[0.0009]		
Bank/month fixed effects	Yes	Yes		
Industry fixed effects	Yes	Yes		
Observations	620,325	620,325		

a Table VI: Default rates and interest revenues

Table VII
Effect of Bank External Financing Expansion on Borrower Debt: Firm Level and Within-Firm Estimations of the Reduced Form of Change in Firm Debt on Changes in Expected External Finance

Columns 2 through 7 present estimated coefficients from a specification of the form:

$$\ln(Debt_{ikt}) - \ln(Debt_{ikt-1}) = \alpha_k + \alpha_i + \alpha_t + \psi(\ln F_{ikt} - \ln F_{ikt-1}) + \chi(\ln(Debt_{ikt}) - \ln(Debt_{ikt-1})) + \epsilon_{ikt},$$

where the left-hand side variable is the debt growth of firm k with bank i at month t . On the right-hand side, F_{ikt} are sources of capital net of reserve requirements (instrumented with expected external financing, $\hat{F}_{ikt}$). α_k , α_i , and α_t are firm, bank and, month fixed effects, respectively; and $Debt_{ikt-1}$ represents the sum of all the bank debt held by firm k , excluding bank i . Column 1 presents the coefficients of the same specification but with the information aggregated at the firm-month level. Thus, the estimations of column 1 do not allow for error clustering at the bank level, bank fixed effects, or the inclusion of $Debt_{ikt-1}$ as a control. Columns 1 and 2 show the estimates obtained using the entire sample of firms. Columns 3 and 4 show the estimates obtained using the subsample of firms that borrow from more than one bank in the sample, which is required for the inclusion of $Debt_{ikt-1}$ as a control. Columns 5 and 6 are estimates of the subsamples of firms that are not eligible and eligible to the MYPES program, where only firms with less than \$200,000 in annual sales and 20 workers are eligible. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

Unit of Observation Borrower Sample Dependent Variable: Debt (logs, first differences)	Loan					
	Firm All (1)	All (2)	With Two or More Lenders		Noneligible to the MYPES (5)	Eligible to the MYPES (6)
	(3)	(4)	(3)	(4)	(5)	(6)
Sources of capital	0.0025***	0.0026**	0.0034**	0.0034**	0.0037***	0.0012
(logs, first differences)	[0.0001]	[.0012]	[.0015]	[.0015]	[.0003]	[.0014]
Debt with other banks						
(logs, first differences)						
Borrower FE	Yes	Yes	Yes	Yes	Yes	Yes
Bank FE	No	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
R-squared	0.11	0.12	0.06	0.06	0.12	0.08
Observations	3,088,481	3,275,779	371,512	288,741	3,226,692	49,171
Error clustering	Firm	Bank	Bank	Bank	Bank	Bank

b Table VII: Borrower debt response

6.6 Table VIII and Table IX: Testing identifying assumptions

Read:

- Table VIII tests whether actual or expected program financing changes banks' marginal cost of capital or risk profile.
- The dependent variables are interbank debt rates and subordinated debt rates.
- The coefficients on actual and expected program financing are statistically insignificant.
- This supports the assumption that program funding was inframarginal and did not change the marginal cost of capital.
- Table IX tests whether program financing is correlated with lagged deposit growth or cash flows.
- Actual program financing is negatively related to some lagged deposit-growth variables, suggesting endogenous participation.
- Expected program financing is not significantly related to lagged deposits or cash flows, supporting its use as an instrument.

Economic message: These tables support the exclusion restriction. The expected-financing instrument is less contaminated by bank-specific liquidity shocks and investment opportunities than actual program participation.

Table VIII
Testing for Changes in Bank Risk Profile: Regression of Changes in Inter-bank Rate and Subordinated Debt Rate on Changes in Actual and Expected External Financing

Each column presents estimated coefficients from a specification of the following form:

$$R_{it} - R_{it-1} = \alpha_i + \alpha_t + \beta \ln(C_{it}/C_{it-1}) + \varepsilon_{it},$$

where R_{it} is either the interbank rate (columns 1 and 2) or the subordinated debt rate (columns 3 and 4) faced by bank i at time t . The rate for each type of debt is calculated dividing the interest rate accrued (from the earnings report at month t) by the stock of debt (from the balance sheet at time $t - 1$). C_{it} is either actual (columns 1 and 3) or expected (columns 2 and 4) program financing of bank i at time t , and α_i and α_t are bank and month fixed effects. All estimations are performed on the reduced sample of banks that received program financing at some time between 1993 and 1999. Heteroskedasticity-robust standard errors are in brackets, clustered at the bank level. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

Dependent Variable	Interbank Debt Rate		Subordinated Debt Rate	
	(1)	(2)	(3)	(4)
Actual program financing (logs, first differences)	-0.074 [0.358]		-0.001 [0.003]	
Expected program financing (logs, first differences)		1.748 [17.42]		0.173 [0.321]
Bank/month FE	Yes	Yes	Yes	Yes
Observations	991	984	993	922
R-squared	0.15	0.07	0.43	0.42

a Table VIII: Bank risk profile and marginal cost tests

Table IX
Testing for Changes in Liquidity and Investment Opportunities: Regression of Actual and Expected Financing Expansion on Lagged Deposit Growth and Cash Flows

Each column presents estimated coefficients from a specification of the following form:

$$\ln(C_{it}/C_{it-1}) = \alpha_i + \alpha_t + \beta_0 \ln(D_{it}/D_{it-1}) + \beta_1 \ln(D_{it-1}/D_{it-2}) + \beta_2 \ln(D_{it-2}/D_{it-3}) + \beta_3 \ln(D_{it-3}/D_{it-4}) + \beta_4 \ln(D_{it-4}/D_{it-5}) + \gamma_0 \ln(CF_{it-1}) + \gamma_1 \ln(CF_{it-2}) + \gamma_2 \ln(CF_{it-3}) + \gamma_3 \ln(CF_{it-4}) + \gamma_4 \ln(CF_{it-5}) + \varepsilon_{it},$$

where C_{it} is either actual (column 1) or expected (columns 2 and 3) external financing from the program of bank i at time t , D_{it} represents deposits, CF_{it} denotes cash flows (income plus depreciation), and α_i and α_t are bank and month fixed effects. Column 2 presents the estimates using the reduced sample of banks that received program financing at some time between 1993 and 1999. Heteroskedasticity-robust standard errors are in brackets, clustered at the bank level. *, **, and *** stand for statistical significance at the 10%, 5%, and 1% level.

	Actual Program Financing (logs, first differences) (1)	Expected Program Financing (logs, first differences)	
		Program Banks (2)	All Banks (3)
Deposits _{t-1} (logs, first differences)	-0.028 [0.049]	0.02 [0.018]	0.002 [0.002]
Deposits _{t-2} (logs, first differences)	-0.162*** [0.058]	-0.015 [0.011]	-0.001 [0.003]
Deposits _{t-3} (logs, first differences)	-0.094 [0.086]	-0.021 [0.016]	0 [0.004]
Deposits _{t-4} (logs, first differences)	-0.111* [0.055]	-0.027 [0.019]	-0.002 [0.003]
CashFlow _{t-1} (logs)	0.00078 [0.0035]	0.00074 [0.00071]	0.00076 [0.00073]
CashFlow _{t-2} (logs)	0.0033 [0.0035]	0.00070 [0.00074]	0.00055 [0.00040]
CashFlow _{t-3} (logs)	0.0058 [0.0056]	0.00096 [0.00098]	0.00027 [0.00044]
CashFlow _{t-4} (logs)	0.0049 [0.0071]	0.00071 [0.00074]	0.00019 [0.00044]
Bank/month FE	Yes	Yes	Yes
Observations	1001	1003	5818

b Table IX: Liquidity and investment-opportunity tests

7 Short summary

- The paper studies whether local bank financial constraints reduce firm access to external finance.
- The empirical setting is Argentina's MYPES program, which allocated government-backed external financing to banks using a formula.
- The paper constructs expected external financing from formula exposure and wave size, using it as an instrument for changes in bank sources of capital.
- The preferred estimates use the final four program waves, which were driven by an administrative deadline rather than banking-sector liquidity needs.
- Banks expand lending by about \$0.66 immediately for each additional dollar of external finance.
- The cumulative response averages more than \$1.2 of lending per dollar of finance over two years.
- Marginal loans issued during lending expansions are not more likely to default, and interest revenue does not fall.
- Borrowers' total bank debt rises when their lenders receive financing shocks, indicating that other banks do not fully replace the constrained bank's credit.
- Identification tests show that expected financing is not related to lagged deposits, cash flows, interbank rates, or subordinated debt rates.
- The paper concludes that intermediary financing frictions can cause underinvestment and reduce aggregate credit supply.

8 Conclusion

8.1 Core conclusion

Paravisini shows that local banks in Argentina were financially constrained and that relaxing those constraints increased credit supply. The lending response to external finance is economically large, rapid, persistent, and amplified. Importantly, the marginal loans funded by the shock do not look worse on either default or interest-revenue measures. This supports the conclusion that constrained banks had profitable lending opportunities that they could not fund before receiving additional external finance.

8.2 Economic intuition

The economic intuition is that local banks hold private information about their borrowers and investment opportunities. If these banks cannot raise enough external capital, profitable relationship loans remain unfunded. Other banks, especially larger or foreign banks, may not be able to step in because they lack borrower-specific information or because switching lenders is costly. When the constrained bank receives financing, it can expand lending to borrowers who were previously rationed.

8.3 Incremental contribution revisited

The paper advances the lending-channel and financial-intermediation literature in three ways. First, it uses a formula-based funding shock rather than endogenous deposits or monetary-policy variation. Second, it directly tests marginal loan quality to distinguish underinvestment from free cash flow. Third, it links bank funding shocks to borrowers' total debt, showing that local bank constraints affect aggregate credit availability rather than merely reshuffling credit across banks.

8.4 Implications for banks, borrowers, and policy

The findings imply that bank balance sheets matter for real credit allocation. Policies that improve constrained local banks' access to capital can increase credit supply and reduce borrower rationing. At the same time, the paper also cautions that program design matters: formal targeting rules may be ineffective if banks can relabel loans. The broader implication is that the health of relationship lenders can be central for small and opaque borrowers.

8.5 Limitations

The study is based on an emerging-market setting with a specific government lending program. The estimates are local to banks affected by MYPES and to a period before Argentina's 2001 crisis. The default tests are measured during normal times, so they may not fully reveal loan quality under severe aggregate stress. The paper also cannot observe all possible nonbank sources of borrower finance. Even so, the combined bank-level, loan-level, and borrower-level evidence provides a strong case for intermediary underinvestment.

8.6 Takeaway

The main takeaway is that local bank financial constraints can bind in economically important ways. When those constraints are relaxed, banks do not simply reshuffle their balance sheets or waste funds on bad loans. They expand profitable lending, and borrowers obtain more total credit. The paper turns the broad idea of a bank lending channel into direct evidence that constrained intermediaries can reduce firm access to external finance.